

2023



## Using AWS Services for Windows to Support Life-Sustaining Clinical Trials with Suvoda

Learn how Suvoda built a resilient infrastructure for its clinical trial technology platform using AWS.

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**Virtually 24/7 uptime**

for critical infrastructure



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in recovery point objective

**5x accelerated deployment**

**Up to 5x reduction**



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## Overview

As clinical trials become increasingly complex, expand to more regions across the globe, and research growing therapeutic areas, [Suvoda](#) is committed to solutions that are supported by the most advanced, secure, and efficient technologies available. Because Suvoda delivers clinical trial technology solutions that support mission-critical moments in life-sustaining trials, it is imperative that clinical trial study teams have quick access to Suvoda's applications around the clock.

Suvoda decided to migrate all of its offerings to Amazon Web Services (AWS) to benefit from the resiliency that AWS offers across multiple [Regions and Availability Zones](#). To build an advanced infrastructure designed to support increasingly complex clinical trials, Suvoda used a suite of AWS services geared toward the deployment of Microsoft SQL Server workloads. As a result, the company significantly improved its disaster recovery and could better serve sponsors and patients who depend on Suvoda products in their clinical trials.



## Opportunity | Using AWS to Migrate Mission-Critical Workloads to the Cloud for Suvoda

Suvoda was founded in 2013 in Philadelphia and specializes in trials for oncology, rare diseases, and the central nervous system. It has supported over 1,000 clinical trials in 80 countries. Its technology automates and manages mission-critical processes that affect clinical trial patients. One of its core offerings, Interactive Response Technology (IRT), helps clinical trial sponsors and sites manage the randomization of patients and logistics. "Our commitment to excellence, in both our technology and our services, is what set us apart," says Priyanka Sharma, Suvoda's vice president of software engineering. "We want clinical trial study teams to spend less time on administrative processes and more time on research and patient care."



Suvoda's applications used .NET Framework, powered by servers in two Romanian data centers, and needed a near-real-time disaster recovery strategy for its SQL Server databases. Additionally, Suvoda required a scalable infrastructure that met stringent industry compliance standards while remaining available virtually 24/7 to meet patients' needs. "The patients who depend on our systems tend to be very sick, so the urgency is high," says Sharma. "Because our system controls drug dispensation, reliability and stability are paramount."

Suvoda collaborated with the AWS team to build a resilient infrastructure in the cloud and migrated all production applications to the new infrastructure in two quarters. Engineers used AWS whitepapers and [The Amazon Builders' Library](#) as sources of ideas for the infrastructure. They also participated in AWS Immersion Days—in-person workshops in which AWS solutions architects walked engineers through AWS offerings. "This was a great opportunity for our engineers to engage with AWS experts, exchange ideas, and learn how the industry is trending and how to achieve successful outcomes using AWS," says Cosmin Ilie, Suvoda's senior director for site reliability engineering.

## Solution | Reducing Recovery Point Objective by a Factor of Four Using AWS Elastic Disaster Recovery

Working alongside AWS experts, Suvoda decided to replatform its highly critical systems, using automation and foundational AWS compute services to build resiliency. For highly available, high-performing storage, it uses [Amazon FSx for Windows File Server](#), which provides fully managed file storage built on a Windows server that can be accessed over the industry-standard server message block protocol. Applications that require fast, low-latency, on-premises access to group shares based on this protocol use [Amazon FSx File Gateway](#).

Suvoda lifted and shifted its internal-facing, noncritical applications to the cloud using [AWS Application Migration Service](#), which minimizes time-intensive, error-prone manual processes by automating the conversion of source servers to run natively on AWS.

Clinical trial teams need reliable network connections to Suvoda's applications. To accelerate traffic, Suvoda uses [AWS Global Accelerator](#), a networking service for improving the availability, performance, and security of public applications (under the AWS [Shared Responsibility Model](#)). "Using AWS Global Accelerator, we are able to provide excellent service to our clients nearly anywhere," says Sharma. "By reducing latency, we are ready to support the changing landscape and exceed the needs of our customers."



Suvoda invested considerable effort in establishing robust disaster recovery capabilities and improving the recovery point objective, including hosting in multiple Availability Zones for high availability and cross-Region deployments for disaster recovery. It deployed mission-critical databases using two SQL Server nodes on [Amazon Elastic Compute Cloud](#) (Amazon EC2), which provides secure and resizable compute capacity for virtually any workload. The two SQL Server nodes were deployed in multiple Availability Zones, with another node in a separate AWS Region for disaster recovery.

Besides the built-in replication capabilities of core AWS services, Suvoda uses [AWS Elastic Disaster Recovery](#), which minimizes downtime and data loss with fast, reliable recovery of on-premises and cloud-based applications using affordable storage, minimal compute, and point-in-time recovery. Thus, Suvoda reduced its recovery point objective by a factor of four and cut its recovery time objective to minutes in case of disaster. Overall, Suvoda maintains virtually 24/7 uptime.

Rather than creating new instances manually, Suvoda uses [AWS Auto Scaling](#), which monitors applications and automatically adjusts capacity to maintain steady, predictable performance at low cost. To further automate IT operations, Suvoda relies on [AWS Systems Manager](#), a secure, complete management solution for resources on AWS and in multicloud and hybrid environments. And using infrastructure-as-code tools such as Terraform, the company reduced the time to spin up new environments by five times. Suvoda further improved performance and sped deployments by containerizing its applications using [Amazon Elastic Kubernetes Service](#) (Amazon EKS), a managed Kubernetes service to run Kubernetes on AWS and on-premises data centers.

## Outcome | Planning Next-Generation Products on AWS

Suvoda uses .NET Core for its next-generation products: the clinical outcomes assessment tool, Suvoda eCOA, and the informed consent tool, Suvoda eConsent. The company also plans to build future applications natively on AWS and to use AWS for all nonproduction workloads by the first quarter of 2024. “Our journey with AWS has focused on providing an infrastructure that can effectively host all of our products,” says Sharma. “On AWS, our infrastructure supports the increasing complexity of clinical trials so that our customers can focus on life-saving research.”

## About Suvoda

Suvoda is a global clinical trial technology company dedicated to supporting complex studies for life-sustaining therapies, specializing in oncology, rare disease, and central-nervous-system trials.

## AWS Services Used

### AWS Elastic Disaster Recovery

AWS Elastic Disaster Recovery (AWS DRS) minimizes downtime and data loss with fast, reliable recovery of on-premises and cloud-based applications using affordable storage, minimal compute, and point-in-time recovery.



recovery.

[Learn more »](#)

## AWS Application Migration Service

AWS Application Migration Service minimizes time-intensive, error-prone manual processes by automating the conversion of your source servers to run natively on AWS.

[Learn more »](#)

## AWS Global Accelerator

AWS Global Accelerator is a networking service that helps you improve the availability, performance, and security of your public applications.

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## Amazon FSx for Windows File Server

Amazon FSx for Windows File Server provides fully managed shared storage built on Windows Server, and delivers a wide range of data access, data management, and administrative capabilities.

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